



ANDREW (KOHLER) WOOD III

MECHANICAL ENGINEER, EIT

Entry-level mechanical engineer with hands-on experience taking CAD designs from concept through prototype and fabrication.

CONTACT

- 📞 208-720-4635
- ✉ akwood1@mines.edu
- 🌐 kohler.solokit.app
- 📍 Lakewood, CO

CERTIFICATIONS

Engineer in Training EIT

In Progress application submitted Jan 2026

FE Mechanical Exam

Passed October 2025

Six Sigma Yellow Belt

Awarded by Colorado School of Mines May 2025

SKILLS

CAD

SolidWorks parts/assemblies/drawings, GD&T basic, tolerance stack-ups basic, DFM/DFA

Simulation

FEA linear static, buckling, CFD SolidWorks Flow Simulation—basic internal flow/heat transfer

Manufacturing

CNC router/mill/laser/waterjet/plasma, MIG welding, metal/wood fabrication, FDM/SLA 3D printing

Methods/Tools

FMEA, Scrum sprints, backlog, standups

Programming

MATLAB, Python, C++, Arduino, LaTeX

EXTRACURRICULAR & INTERESTS

ASME member · Woodworking · Certified Rescue Diver · Advanced skier · Jewelry fabrication

EXPERIENCE

Prototyping & Fabrication Projects

March 2025 – Present

LABRIOLA INNOVATION HUB, MINES • GOLDEN, CO

- Designed and fabricated woodworking items cutting boards, rolling pins, record cabinet using SolidWorks models/drawings; executed long miter joints requiring tight fit-up and consistent tolerances.
- Operated ShopBot CNC router and shop equipment; selected grain orientation and specified food-safe adhesives/finishes; improved dimensional consistency through refined setups.

Senior Capstone Project – Adaptive Bass Guitar

Aug 2024 – May 2025

COLORADO SCHOOL OF MINES • GOLDEN, CO

- Redesigned the plucking mechanism to eliminate failure modes from flexible/unstable connections; implemented 3D-printed paddle interfaces to improve reliability in repeated use.
- Designed a player stabilization system shoulder strap + hip strap to counteract playing-induced torque and replace a tripod-based neck support, improving usability and reducing setup inconvenience.
- Modeled assemblies/mechanisms in SolidWorks and produced a detailed assembly manual.
- Ran work in 2-week Scrum sprints; tracked risks/mitigations with FMEA.

Mechanical Design Assistant

Summer 2022

JG WORKS • HAILEY, ID

- Fabricated mono-ski hardware for Special Olympics athletes.
- Assisted in designing modular office space using 3-axis CNC cut plywood sheets.

Fabrication Intern

Summer 2021

PHRED'S FABRICATION • KETCHUM, ID

- Assisted with layout, fabrication, and installation of steel railings for residential projects.

EDUCATION

Colorado School of Mines

Aug 2021 – Aug 2025

B.S. Mechanical Engineering
GOLDEN, CO

Macy-Naszkowski School of Art Music

Oct 2022 – Present

Additional Training: Classical Piano
LAKEWOOD, CO

